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FORUM

Finance, Land and Labour

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The dynamics of contemporary capitalism have empowered the role and influence of finance within the realm of agriculture. In response, agri-finance research has focused on the extension of global finance's investment chains and how parts of the agricultural sector—mainly farmland—are transformed into financial assets, where multiple sources of capital seek to make gains. To do so, scholarship on 'finance going farming' has directed attention to the variety of financial actors (i.e., pension, endowment and private equity funds, insurance companies and investment banks); their motives to invest; the variegated mechanisms deployed to reformat farmland and agricultural production for financial purposes; and the increasing power of shareholders to shape productive and distributive decisions. While this literature has advanced our understanding of how finance makes its way into agriculture, within agrarian studies, these processes and dynamics raise important conceptual and methodological challenges about how to centre financialization in dynamics of agrarian change. In this exchange, our contributors consider how contemporary trends in agri-finance demand us to rethink relations of production, property and power and processes of accumulation. Key questions that the forum addresses include how and to what extent is finance connected to the restructuring of capital and its modalities of accumulation in agrarian settings? What ties does financialization have to changes in labour regimes? How does it affect productive capital and its associated relations of power? This Forum is part of the 25th Anniversary Forums¹, following 'How is climate change changing agrarian studies?' (Paprocki et al. 2025), 'Challenging agroecology—Promise and pitfalls for agrarian studies' (McKay et al. 2025) and 'What is the value of value for agrarian studies' (Akram-Lodhi et al. 2025).

1 | Response 1: Samuel Frederico

The growing hegemony of finance has been reshaping the mechanisms through which land and nature are appropriated. In global agriculture, institutional investors—such as pension funds, insurance companies, private equity firms and sovereign wealth funds—have become central actors in reorganizing the circuits of accumulation by shifting the focus from production to asset valorization and the extraction of territorialized rents (Cotula 2012; Isakson 2014; Clapp and Isakson 2018). This text contributes to the debate by examining how land is increasingly treated as a financial asset through the articulation of rentier and speculative logics, suggesting that asset managers play a pivotal role as strategic intermediaries who coordinate investment flows, mediate relationships between financial investors and local actors, and embed financial rationalities into the operational logic of farming enterprises (Clapp and Isakson 2018).

Although often grouped under the label of 'financial capital', institutional investors differ considerably in their profiles, time horizons and modes of territorialization (Knuth 2015; Ouma 2020). Among them, asset management firms have emerged as key protagonists in the financialization of the global economy (Christophers 2023). Acting as intermediaries, managing pooled resources from pension funds, insurance companies, endowments and high-net-worth individuals, they allocate capital across asset classes to maximize returns, shaping global investment flows and corporate strategies in the agrifood sector.

These dynamics operate through complex trans-scalar arrangements that obscure ownership structures and hinder legal

accountability. The ‘architecture of invisibility’ shields asset managers from the social and environmental consequences of their investments, impeding the identification of both investors and land assets. At the same time, such arrangements reveal that financialization is neither de-territorialized nor purely abstract, but materializes through specific actors, institutions and territorial configurations. To address these transformations, this intervention analyzes (1) the conversion of land into a financial asset and its implications for property relations; (2) the role of asset managers in the trans-scalar governance of farmland, coordinating capital flows and shaping land markets through financial rationalities; and (3) the architecture of invisibility and trans-scalar arrangements that conceal ownership patterns while facilitating the incorporation of peripheral territories into global circuits of capital accumulation.

1.1 | Land as Financial Asset and Property Relations

Although the relationship between finance, land and agriculture is not new (Ouma 2020), it has recently reshaped the traditional categories of agrarian political economy through novel forms of assetization, investment and territorial control—fundamentally altering property relations, production dynamics and power structures within circuits of capital. Financialization articulates and shifts the central axis from production to asset valorisation and the extraction of territorialized rent. Land, historically conceived as a means of production and the basis of landed rent, is increasingly treated as a financial asset (Ouma 2016; Visser 2017; Ducastel and Anseeuw 2017), susceptible to tokenization, securitization and trade on digital platforms (Reisman et al. 2025).

As Clapp and Isakson (2018) emphasize, the boundaries between productive and financial capital are increasingly blurred. Even when involving productive activities, agriculture under financial control tends to prioritize and combine rentier and speculative logics over productive engagement (Gunnøe 2014). Financial funds combine the appropriation of land rent with the management of agricultural production through subsidiaries or lease contracts (Fairbairn 2014, 2020; Knuth 2015). By generating expectations future returns, value is deferred into the future, and land is transformed into fictitious capital (Harvey 2006) that is, a claim on future income that is abstract and disconnected from its social and ecological materiality.

In this context, distinguishing between rentier and speculative logics of land control remains essential. The rentier logic involves long-term strategies aimed at securing territorialized rents, typically through land leasing, regular income flows and indirect yet sustained productive engagement. The speculative logic, in contrast, seeks short- to medium-term capital gains by acquiring and reselling farmland assets, often with little regard for local integration or productive use. However, in practice, these logics are not mutually exclusive. Many financial actors simultaneously adopt rentier strategies to ensure predictable returns while positioning their land holdings for speculative appreciation. It is precisely this combined deployment of rentier and speculative rationalities that characterizes the financialization of agrarian frontiers today, reshaping land control through both ongoing extraction and anticipatory valorization.

Although grouped under the umbrella term ‘financial capital’, institutional investors pursue distinct strategies (Daniel 2012; Knuth 2015; Clapp and Isakson 2018). Their operations in Brazil’s agricultural frontier are perhaps emblematic of the diversity of financial capital in the agrarian sector (Frederico 2019). There is a wide range of actors, from those seeking rapid capital appreciation—such as private equity firms—to those aiming for long-term territorial embeddedness, including Chinese, Japanese and other Asian corporations. While the former typically adopt a more speculative approach centred on the purchase and resale of assets (in this case, farmland), the latter follow long-term strategies tied not only to food security agendas but also to broader geopolitical interests of their countries of origin, as is the case with Chinese state-backed investments.

There are also investors that fall somewhere between these two poles. Notably, pension and endowment funds from the United States and Western Europe tend to adopt medium to long-term investment horizons. In addition, since the early 2000s, there has been a noticeable migration of capital originally specialized in urban real estate into agriculture—exemplified by partnerships between Argentine and Brazilian capital through firms such as Brasilagro (Gras and Nascimento 2017). It is important to emphasize that international financial capital rarely operates in isolation. In most cases, it seeks partnerships with domestic investors, national agribusiness firms and local producers (Frederico and de Almeida 2019). As one of the largest agricultural investment funds operating in Brazil’s frontier regions stated, success relies heavily on knowledgeable and experienced people on the ground². This underscores the importance of a trans-scalar analytical approach that connects the owners of capital with the actors embedded in specific territories—such as large landowners and agricultural firms—while also accounting for the mediating role played by asset managers in orchestrating these multilevel interactions.

1.2 | Asset Managers as Strategic Intermediaries in the Trans-Scalar Governance of Farmland Investments

A key dimension of the financialization of agrifood systems lies in the role of asset managers as central agents of capital restructuring at a global scale. By shaping investment trajectories and controlling substantial capital flows, asset managers have positioned themselves as central actors in global value chains and in the governance of strategic sectors (Christophers 2023). The 10 largest asset management firms—including BlackRock, Vanguard and State Street—collectively manage over US\$50 trillion in assets, underscoring their extraordinary influence in the contemporary financial system (Phillips 2022).

Asset managers have significantly expanded their footprint within the global agrifood system. As Clapp (2019) notes, this transformation has not only restructured ownership patterns—through common ownership of competing firms—but also aligned corporate governance practices around the imperative of shareholder value maximization. In agriculture, asset managers typically do not engage in direct production. Instead, they acquire farmland through specialized investment vehicles, lease it to local operators or subsidiaries and extract value through

both rent-based income and the appreciation of land as a financial asset. These strategies enable them to act as absentee land controllers, embedding financial logic into agrarian frontiers through rentier and speculative practices.

The distinction between conventional agribusinesses and those controlled by asset managers lies less in their day-to-day operational practices—such as optimizing land use or reducing production costs—and more in the structural logic, time orientation and governance structures that shape their strategies (Clapp 2019). Although both types of companies are involved in commodity production and seek to maintain productive capacity and resilience, companies controlled by asset managers operate under a distinct financial rationale. Their main orientation is to maximize asset value and guarantee returns for distant shareholders, which generally implies shorter investment horizons, enhanced liquidity requirements and performance metrics based on financial abstraction and portfolio optimization. Governance in these companies is directed by asset managers acting on behalf of large institutional investors, with decisions grounded in global financial metrics rather than local socio-environmental conditions. As a result, the logic of accumulation becomes increasingly disconnected from the specificities of place, reinforcing speculative and mobile forms of capital deployment across agrarian landscapes.

This restructuring, however, is not confined to internal corporate transformations. It also entails new modalities of capital territorialization. Asset managers expand their influence by operating through subsidiaries, joint ventures, partnerships with local firms and the creation of farmland-specific investment vehicles. These organizational forms allow them to embed financial strategies into agrarian frontiers while maintaining operational flexibility and minimizing political and legal exposure. In doing so, they deepen the disembedding of agrarian production from local dynamics, while reconfiguring land not merely as a factor of production, but as a financial asset circulating through transnational investment chains.

The speculative nature of asset managers' strategies and their detachment from the territories in which they operate are perhaps best illustrated by the statements made during an interview we conducted with the CEO of a large agribusiness company operating in Brazil. The conversation took place in a smart glass building in São Paulo's financial district—more than 2,000 km away from the company's farms. Immediately after the presentations, the executive bluntly commented: 'I don't know the difference between a corn plant and a soy plant', in reference to his lack of knowledge about agricultural production aspects. He added that his job was simply to 'dress up the bride', a metaphor for making the company look attractive to investors, to guarantee a rapid rise in value before an anticipated 'divestment'—i.e., a complete sale of the company—within 2 years. This anecdote sums up the financialized logic that guides asset managers: The central concern is not the long-term productive engagement with the land, but the potential to extract capital gains through strategic exits, with little regard for local agrarian realities.

In this sense, it seems necessary to deepen the analysis of how asset managers (re)construct 'ethical' narratives to attract investors (Parfitt 2024), deploy metrics to transform agricultural

commodities, land and natural resources into financial assets (Ouma 2016, 2020), and actively engage in regulatory agendas alongside states and supranational bodies in defence of their interests. While the broader phenomenon has been widely studied, there remains a lack of detailed investigation into their specific operations in agriculture—particularly in the Global South. Analyses that focus on the territorial impacts of these financial operations, the role of local intermediaries and the socio-environmental consequences of disembedding and re-embedding land into global investment circuits would help illuminate the concrete dynamics through which financialization operates on the ground.

1.3 | Architecture of Invisibility and Trans-Scalar Assemblage of Land Control

One of the main empirical challenges faced by research on the financialization of agriculture lies in rendering visible the actors situated at both ends of capital flows (Ouma 2023). Upstream, the task involves tracing the origins of capital—identifying those who finance the agricultural funds managed by asset managers. Downstream, the challenge is to map the landholdings controlled by these funds and analyse their territorial impacts. In both directions, financial actors contribute to an 'architecture of invisibility' that complicates the identification of both investors and land assets. Asset management firms often withhold the names of fund participants and establish multiple legal entities with distinct registrations and names, even when these belong to the same corporate group, thus complicating the tracking of landholdings under their control (Spadotto et al. 2020).

Analysing these dynamics requires methodological tools capable of interpreting the spatial distribution and concentration of capital (Hughes-McLure 2022). In this regard, the 'following the money'³ approach has become central to the field of financial geography. Through this method, researchers aim to map capital circuits and the actors involved in financial and commercial transactions. However, obtaining information about the origin of financial flows remains a significant challenge. Asset managers frequently impose barriers to traceability, deliberately concealing the identity of those who capitalize the funds they administer.

This opacity becomes particularly evident in the case of one of Brazil's largest agricultural enterprises, which controls over 300,000 ha in frontier regions. The company is owned by a São Paulo-based asset management firm specializing in so-called alternative investments, including urban real estate, farmland, planted forests and privately held companies. According to information disclosed by the firm itself, the establishment of the agricultural company was aimed at attracting foreign institutional investors seeking exposure to real assets in Brazil. Currently, the firm manages approximately 20 financial funds with assets exceeding US\$2 billion, sourced from around 50 foreign institutional investors, including pension funds, endowments and sovereign wealth funds—50% from the United States, 35% from Europe and 15% from the Middle East and Asia.

Despite the appearance of transparency, the publicly available information is notably vague and insufficient. It does not provide

detailed data on fund participants, nor does it reveal connections between capital flows and the specific landholdings under control. Disclosure is limited to broad investor categories and geographic regions of origin, while the names of participating institutions and the precise nature of their involvement remain undisclosed. This lack of traceability points to certain analytical limitations of the ‘following the money’ approach, especially in contexts where asset managers employ complex structures that make it difficult to map the connections between financial capital and land control.

Downstream, direct control over land is also obscured by various strategies. Chief among them is the outsourcing of land operations, drawing a parallel with labour outsourcing in certain agricultural sectors, such as sugarcane harvesting, which absolves mill owners of responsibility for degrading labour conditions and cases of forced labour. Similarly, international funds deflect accountability for the impacts associated with land acquisitions, as they are not direct landowners but equity partners in agricultural ventures (Pitta and Mendonça 2016).

This outsourcing mechanism involves the creation of multiple companies with the same administrators, including subsidiaries, to give the impression that landholdings belong to distinct owners. Typically, each farm controlled by the same financial group is registered as a separate legal entity, thereby masking the group’s consolidated control. Many of these properties are acquired through irregular means and affect traditional communities living in surrounding areas. However, the outsourcing structure prevents firms from being held accountable for social and environmental impacts. As shown by Pitta and Mendonça (2016), such entities often engage in intragroup land transactions to stimulate the land market, fuelling speculative dynamics and contributing to the inflation of land values, which in turn reinforces financial interest in land assets.

The proliferation of distinct legal entities serves to obscure land ownership and limit the visibility of the investors involved. Outsourcing structures also enable firms to avoid being classified as direct landowners, thereby bypassing legal responsibilities regarding land-related impacts.

Beyond the challenge of identifying the investors who capitalize asset management funds, another critical issue is understanding how asset managers locate and invest in land situated in marginal areas of the Global South. This requires a trans-scalar analytical perspective that connects the origin scale—linked to the financialization of the global economy and the growing power of asset managers—with the implementation scale, where concrete forms of cooperation, conflict and resistance emerge in targeted regions (Frederico and de Almeida 2019).

The trans-scalar nature of financialized land control is particularly evident in Brazil’s agricultural frontier. To access land markets, financial investors establish relationships across multiple scales—from global corporations and supranational institutions to local land grabbers, politicians, farmers and large landowners—while also relying on the crucial role of the state in land governance, credit provision and infrastructure development.

In the previously mentioned case, following the formation of the funds and the capital raised from institutional investors, the asset manager established the agricultural company responsible for land acquisition and conversion for production. To accelerate the expansion of its land portfolio and optimize identification, negotiation and management processes, the firm entered into a strategic partnership with two families of Brazilian landowners with holdings in key frontier regions. This case illustrates a recurring dynamic in contemporary land financialization: the articulation of actors operating at multiple scales. The process involves, on one side, globally positioned institutional investors—such as pension funds, endowments and sovereign wealth funds—and, on the other, local producers with specific territorial knowledge. At the intermediary level, asset management firms and their operational subsidiaries act as key mediators in the territorial embedding of financial capital. This constitutes a trans-scalar network of cooperation and intermediation that links global financial centres to peripheral zones of agricultural expansion in the Global South.

2 | Response 2: Stefan Ouma

Almost 15 years have passed since Tania Li’s seminal paper, ‘Situating Labour in the Land Grab Debate’ (Li 2011), was published. With over 1,200 citations on Google Scholar, one might expect us to have arrived at a profound understanding of the impact of the global land grab on labour regimes and capital–agricultural labour relations across diverse geographies.

From a broader perspective, the answer to that expectation is ‘yes’. Indeed, a growing body of literature has examined how large-scale investments in agriculture reshape the spheres of production, reproduction and agricultural labour markets at large. This often includes contract-farming arrangements, where these spheres are closely intertwined, as household labour is frequently mobilized—often in cheapened or unpaid forms—to subsidize contract-based agricultural production.

However, once we centre finance in a more substantial way by looking into financialized farmland or agricultural operations—something Li did not do in her seminal paper—‘labour’ still remains a lacunae. Major empirical works on capital placements in agriculture,⁴ such as Fairbairn (2020), Dixon (2023), Langford (2023) and my own (Ouma 2020), say very little about labour, class relations and labour struggles (for a constructive critique of this, see also Sturman 2024). Cochet’s (2018) work on the labour-value nexus in large-scale farming is one of the few major contributions on the topic, yet it has received only 27 citations since its publication. Assuming that large-scale agriculture is defined by a separation of capital and labour, unlike the ‘family farm’, the Cochet’s paper focuses on different types of large-scale farming in Ecuador, Ethiopia, Ukraine and South Africa, in order to ‘study the way value added is distributed in order to compare the shares going to labour compensation, capital (investors and financial institutions ...) and other actors (rental payments for land, taxes or duties paid to the state)’ (ibid.: 18). Although it is not solely focused on cases where agricultural investment funds drive the expansion of large-scale farming, it is still a useful piece due to its ambition to quantify how value is distributed between capital and labour.

At the same time, empirical and research-related developments have further complicated the labour question, requiring us to acknowledge the multiple ways to think through the finance-land-labour nexus. In this intervention, I will argue that the financialization of farmland and agricultural production can be connected to labour issues in at least six promising ways. During this, I will stick to my own line of research: the placement of equity capital in production. That said, finance's penetration of agriculture also includes the classic vehicle of 'credit' (Green 1987; Le Heron 1991; Green 2022) as well as other stages of the agricultural value chain (Clapp and Isakson 2018; Burch and Lawrence 2013).

The first approach discussed here follows on from Li's (2011) original work. This approach directs our attention towards the dynamics of inclusion and exclusion arising from the large-scale placement of capital in agriculture. What labour relations and practices are established on assetized farms? What racialized, gendered and classed legacies do these rely on? How do these arrangements differ from those on farms run as plantations or large-family farms with a long-term farming intention, where rentierism and speculation play less of a role? How do such placements shape labour markets over longer periods of time? In my own research, I demonstrated that investors targeting farms in Aotearoa New Zealand did not need to develop new methods of labour exploitation. Instead, they could build upon three decades of market liberalisation and 'farm professionalization', which resulted in the development of a flexible labour market system (Stringer 2016). This system often relied on South/Southeast Asian or Pacific Islander workers, who were hired through labour market intermediaries, as seen in the dairy industry. Alternatively, farms could utilize external contractors, or 'teams for hire', for specific services, as observed in the fruit industry.

However, what farm managers and investors pressured by the drive for shareholder value do is think more deeply about labour incentivization and monitoring. This results in a practice that I call 'dashboard farming' (Ouma 2020, 141). For instance, the Map of Agriculture project⁵ emerged from the desire of a New Zealand agri-focused asset manager to create a 'Bloomberg for agriculture' so that both the scouting for farms and their management become better aligned with the data-centred work of institutional investors and asset managers.

This insight comes with two caveats. First, given the wide variety of investment geographies, these findings cannot easily be generalized, and we must always consider the finance capital-labour nexus within its specific historical and geographical context. Secondly, critics might argue that finance underpins *all* investments in agriculture, whether they are classified as ordinary, so-called strategic agribusiness investments (e.g., Del Monte opening a new pineapple farm), or as instances of assetization, where new players target agriculture solely for the purpose of asset creation. From this perspective, the emerging literature on large-scale land grabs addressing labour issues might be considered a step in the right direction. Indeed, there are cases where both logics cannot be easily separated (Cochet 2018). However, we need to operate more carefully and adopt a distinction between the logic of investment—which might shape classical agribusiness decisions—and that of capital placements

(see *endnote 4*), where financiers acquire an asset with the explicit intention of speculation and rent-seeking over a relatively short period of time. We need more research that accounts for this distinct logic and thus examines in detail the impact of a focus on financial objectives, metrics and narratives on on-farm labour relations and labour markets, and the mechanisms through which this occurs. At the same time, we need to consider the exclusionary dynamics induced by large-scale capital placements in farmland in particular. What happens to farmers (as well as fisher folk, hunters, beekeepers, nut collectors and pastoralists, for example) who have been dispossessed and displaced by such placements? What happens to the domain of reproductive labour, as is brilliantly illustrated in great empirical detail in Chung's work on Tanzania (Chung 2024)? Although we have seen a lot of good work published in this field in recent years, this work has usually not built a more solid bridge between inclusion–exclusion dynamics and the fundamental logics of capital markets.

A second way of thinking about the relationship between finance, land and labour is to prioritize social reproduction, which, as of more recently, has become a more central concern in the land grab debate and critical agrarian studies at large (Chung 2017; Ossome 2022; Shattuck et al. 2023; Wolford 2021; see also the contribution of Alessandra Mezzadri to this first instalment of this forum). However, it needs to be more explicitly tied to the question of how financialization/assetization dynamics reshape the land–social reproduction nexus. Social reproduction becomes relevant in relation to female farm (or factory) workers, raising the question of how corporate financialization logics shape their lived experiences at the intersection of wage and the household sphere. However, it could also be extended to include the female spouses and children of farmers who are recruited as cheap labour in contract farming, as demonstrated in our own research (Iddrisu et al. 2002), building upon an established tradition that goes back to the heydays of the modes of production debate (Carney 1988). Lastly, finance and social reproduction might be connected via the issue of 'credit' rather than equity. This is superbly demonstrated by Green and Bylander (2021) in their work on microfinance, household indebtedness and distress land sales in Laos. Like the realm of production itself, both aspects might be sites of resistance, and the 'counter-topographies' (Del Rio 2024) emanating from this hold valuable lessons for imagining alternative production and financing methods.

Some readers of this journal may find the third approach challenging, as it replaces the traditional Marxist category of 'labour' with 'work'. Of course, many Marxists are not opposed to broadening the category of 'labour' per se, as evidenced by more recent work on labour as social reproduction or different classes of labour, as shown in this journal (Barbesgaard 2025). However, what might raise a few eyebrows is moving away from the exploited classes—the rural working people, broadly speaking—to encompass all the other players involved in turning agriculture into a financial asset, such as fund managers, placement agents, lawyers, valuers, enterprising family farmers and (internationally mobile) farm managers (Langford 2023). Examining how these 'agents of capital' assemble an agricultural asset through embodied work builds on a praxeological tradition (Boussard 2017) that appears to conflict with Marxist concerns (Fine 2003; Preda 2013). However, adopting the

perspective of the ‘micro-founded political economy of the investment chain’ (Braun 2016, 6), which links the concrete actions of sense-making subjects to more abstract ‘operations of capital’ (Mezzadra and Neilson 2015; Ouma 2016), allows us to salvage the best of both worlds. Andrew Ofstehage’s (2025) recent book, which focuses on how migrant farmers from the US contribute to the creation of ‘soylandia’ in Brazil, is a great example of this.

A fourth way to multiply ‘labour’ within the finance-agriculture context is to introduce non-human labour, taking cues from critical animal and multispecies studies. Financiers, or at least the farm managers they employ, often make conscious decisions about which animal breeds or crops to introduce and how to maximize their output. In my own research into assetized farms in Tanzania and Aotearoa New Zealand, I have observed the impact of factors such as different poultry or dairy breeds or the type of clover planted on the rate at which investment capital is recouped. The concept of non-human labour could also be extended to include the work of microorganisms, which further challenges classic Marxist thinking on labour. This could explain why the seminal book on the subject by Besky and Blanchette (2019) has not yet been cited in any contributions to the ‘land grab literature’ (according to Google Scholar). Similarly, Wolford et al.’s (2024) recent stock-taking article cites an impressive 304 papers but does not mention the words ‘animal’ or ‘non-human labour’ once! While there are limits to how much ‘labour’ we can attribute to animals and other non-human organisms (Marxist Education Project 2024), it would be beneficial to further explore this connection, particularly in light of rapid technological advances that facilitate the manipulation and control of organisms of various sizes, incorporating them into capitalist systems in increasingly profound ways (Guthman and Fairbairn 2024). As we see the further expansion of venture-capital powered platforms and other ‘AG-tech solutions’ (Reisman et al. 2025), this also bears the question of how new forms of precision-farming reshape the labouring capacity of animals, microorganisms and nature at large.

The fifth way to focus on labour in the context of our debate relates to the long-standing interest of those working on the agrarian question. Might capitalist technological progress render certain forms of labour obsolete? Examples of this include the replacement of manual tasks such as milking and cleaning stalls by robots and the replacement of the cognitive work of farm managers and workers by data-collecting tractors and drones. In certain parts of the world, digital technologies and processes of automation are beginning to reshape the domain of agricultural production and its pre- and post-stages to such extents that the classic Mann–Dickinson thesis, which claimed that agriculture cannot be fully subsumed to capitalist logics because of its metabolism and dependence on family labour, needs to be revisited (Mann and Dickinson 1978). When adopting a financialization and assetization optic, new empirical avenues emerge. Financiers tend to prefer lean corporate structures. Reducing capital-intensive costs can improve the appearance of assets on the balance sheet. Similarly, cutting back on human labour, which requires wages and benefits such as health insurance and pension payments and may organize collectively, can also enhance an asset’s value. Digital technologies can aid these processes. For example, capital cost expenditure can be closely

monitored through dashboards, and production processes can be automated (Rotz et al. 2019).

Lastly, we can problematize labour itself as a source of capital that ultimately finds its way into agriculture. This brings us to Mr and Mrs Scheper, two retired German doctors whom I mention in the epilogue of my book. In 2018, they discovered that their retirement provider, the medical chamber of a German state, had invested in a TIAA-CREF fund in Brazil. In the United States, the institution now simply known as TIAA serves teachers, professors and other professionals as a retirement providers. The said fund had been accused of gross social and environmental misconduct (Schwab 2018). As I wrote, ‘Having supported indigenous struggles in Brazil for decades and having personal experience in the field, the Schepers were particularly shocked at what had seemingly happened to their retirement savings’ (Ouma 2020, 181). The Schepers’ experience highlights an uncomfortable question: Where does all the interest-bearing capital flowing into agriculture come from? While the very rich and ultra-wealthy certainly have a stake in this (Bill Gates is currently the largest private farmland owner in the United States!), a significant proportion of this originates from pension funds centred on the working and middle classes. By 2023, an astonishing 960 active funds had specialized in food and agricultural assets (Seufert et al. 2025, 15), managing over US\$150 billion in future pensions for workers in countries with capital market-based retirement systems. While the Schepers are not workers in the strict sense of the word, they are still representative of what I have termed the ‘global return society’—a global socio-material formation in which the reproduction of the better-off people of the Global North (and increasingly the Global South) is tied to the reproduction of finance capital, both ‘at home’ and ‘abroad’ (Ouma 2020, 135; see also Parfitt 2018, for an insightful article). The role of pension funds in the financialization/assetization of agriculture and the underlying politics of reproducing imperial lifestyles of a certain class of labour within transnational ‘value relations’ (Araghi 2003) has received little attention from scholars.

I hope scholars of critical agrarian studies find these reflections useful, and the finance-land-labour nexus will move to the centre of debates for the years to come. Placing labour at the finance-land nexus is not a straightforward affair, as there are multiple entry points for framing the ‘labour question’ in this context.

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3 | Response 3: Emily Duncan

The finance, land and labour nexus continues to form new assemblages and shift over time. To this mix, we must increasingly add digital technologies as a key element of transformation. Digital agriculture is rapidly changing the ways in which farmland is acquired, managed and turned into a financial asset. These tools are used to make land, labour and production more legible and more visible to a range of agrifood actors, leading to new forms of governance. In this contribution, I examine the

growing interconnections between the assetization of land and digital technologies (and their associated data). In turn, I relate these evolving trends to the question of labour, asking: How does the use of new digital tools in assetized and non-assetized contexts affect the nature of agricultural labour?

In considering how to conceptualize the critical role of technology in new forms of accumulation and the changing dynamics of financialization, I examine the coupling of platform capitalism and surveillance capitalism in agriculture. As we continue to theorize about the power of digital technologies to restructure production, ownership and related material relations, there is a need to apply these concepts to the agrarian questions of land and labour. Studies have demonstrated the ways in which these technologies and their associated data are reinforcing corporate power and reshaping agricultural knowledge production, but little work to date has been done to extend the analysis to the assetization of land and changes in labour regimes. I suggest that emerging forms of agrarian platform and surveillance capitalism—with attention to global forces, such as climate change, and regional differences—are key to understanding contemporary processes of agrarian change.

3.1 | New Forms of Capitalism Leading to Agrarian Change

Capitalist modes of accumulation based on the expropriation of land and nature in agriculture now rely additionally on the expropriation of data about land, nature and workers. Farmland becomes dually productive for generating profits through agricultural production and as a site of data production. Digital agriculture provides the tools for this type of profit generation via data; however, farmers may benefit by using agricultural data to improve their own production and farm management practices, while other actors (such as tech or agribusiness firms) can monetize the data and capitalize on the sale of data products.

Platform capitalism is defined by the developments of digital spaces (both hardware and software) designed to enable exchange, or else to create a place for a variety of actors to conduct business (Srnicek 2017). The ability to generate rentier profits has become increasingly important in financialization processes; and platforms, with their monopolistic tendencies, help to facilitate these processes by providing a digital transactional space (Birch 2020; Christophers 2020). While there has been an urban focus in examining platform capitalism, in a key article, Reisman et al. (2025) begin to theorize agrarian platform capitalism by creating a typology of agricultural platforms. Analyses of agrarian platform capitalism align with the critiques of platform capitalism more generally, with some unique features emerging in the agricultural space. While there are some examples of e-commerce or contract marketplaces, there are also e-offerings of data brokerage and data architecture platforms, which rely on vast amounts of digital agricultural data. While not all digital agricultural technologies are built on a platform model, large agribusiness firms are increasingly seeking to develop these sorts of platforms to expand their business offerings. The ability to facilitate farmland investment through platforms with the aim to ‘democratize’ investment into this growing asset class has furthered the process of farmland assetization

(Duncan and Magnan 2023). In this case, data brokerage platforms, like AcreTrader, aim to monetize farm data in order to facilitate a frictionless online farm real estate investment process, making visible and legible individual parcels of land to prospective buyers. Similar to the broader value stories used to promote farmland investment (Fairbairn 2020), platforms assert their moral legitimacy using discourses of feeding the world and environmental sustainability to illustrate the importance of their services (Reisman et al. 2025).

Future scholarship in agrarian studies should endeavour to understand the implications of the platformization of agriculture on labour. Agricultural retailers in rural communities often rely on relationships of trust with their farm clients (Duncan 2018)—how will this workforce be reshaped when interpersonal exchanges are moved to online platforms? In light of the growing use of custom farmwork for hire, how might agricultural labour be reorganized through platforms? Given that urban platforms for ride-share services and food delivery have been incredibly exploitative of labour, there is a need to explore how agrarian platforms could exacerbate inequalities in agrarian labour, especially in the contexts where farms are owned and operated by distant investors (Rotz et al. 2019). The integration of platforms into agricultural processes creates new forms of governance over farming practices, as illustrated by the case of Indonesian fish farmers, who must engage in new forms of work, such as translating their farming practices into the form of data objects legible to the digital platform (Kruk et al. 2025).

In addition to the rise of agrarian platform capitalism, agrarian surveillance capitalism provides new mechanisms for accumulation and allows for novel forms of finance to permeate the processes of agricultural production. Surveillance capitalism, tied to the likes of big tech (i.e., Google, Amazon, Microsoft and Meta), is characterized by the accumulation of vast quantities of behavioural data with the objective of generating value through predictive analytics and ultimately modifying human behaviour in ways that generate even more data (Zuboff 2019). These business models lead to extreme asymmetries in knowledge and power between those generating the data (users) and the corporations controlling the data. There is the need to better understand how new forms of agrarian surveillance capitalism will affect power relations, as both the tech giants and agribusiness incumbent firms develop digital tools that aid in farmland assetization processes (Duncan et al. 2021; Fairbairn and Reisman 2024).

In connecting platformization with agrarian surveillance capitalism, a key mechanism of dispossession is the terms and conditions of data use imposed on farmers. These make technology and platform use—including the ‘decision support’ sought by farmers—conditional on a handover of farm data, in turn modifying the behaviour of farmers (Ruder 2024). For instance, platforms may make recommendations to farmers about specific products to adopt, such as inputs or seeds, that are produced by their parent company. Further, data grabbing is intertwined with land grabbing (Fraser 2019; Hackfort et al. 2024). For instance, through satellite imagery, digital soil and yield maps and other on-farm sensors, massive amounts of data allow potential institutional investors to screen and select potential investment properties more efficiently (Duncan et al. 2022). While farmers’

socio-technical imaginaries include worries that these technologies will decrease their access to farmland, they tend to have a more positive view of the effects of digital agriculture on labour. Specifically, farmers perceive that these technologies will alleviate labour shortages and increase the quality of work (Ruder et al. 2025).

Agrarian surveillance capitalism also has profound effects on labour. Digital technologies allow for the tracking and management of farm labour from a distance, with digital devices providing real-time data about the movement and location of farm machinery and operator efficiency. The surveillance extends to individual farm workers who, in greenhouse production, for example, are required to wear movement devices designed to track how quickly they can pack produce (Doggett et al. 2024). These technologies allow farm owners and managers the ability to monitor and manage their productive assets and workers remotely and in real time, in an effort to generate greater profits.

The digitalization for farms is happening not only in the so-called global north but is increasingly advocated as a solution for farmers in the global south. Digitalization in this context involves technologies like ‘Uber for tractors’ and smartphone apps for weather or marketing information (Tsan et al. 2019; Daum et al. 2021). While they are promoted as solutions to the production challenges facing smallholders (Abdulai 2022), there is currently little empirical evidence that these technologies are able to improve the livelihoods of smallholder farmers as evidenced by the limited use of platforms (Abdulai et al. 2023). Agrarian surveillance capitalism of peasant farmers is not currently widespread, despite its promotion by philanthropic organizations, international institutions and multinational agribusinesses; yet, it poses a threat to farmer autonomy given the effects of these technologies on farmer decision making (Stone 2022; Montenegro de Wit and Canfield 2024).

In considering the shifting patterns of financialization in agriculture due to digitalization, a broad range of actors must be considered. While institutional investors, such as asset management firms, have been able to further their interests through digital tools, other players are driving shifts in farm land ownership. In the Canadian context, studies showed an increase in institutional investment activity from 2002 to 2014 (Desmarais et al. 2017; Mangan and Sunley 2017). Yet, some major players like the Canada Pension Plan have recently divested from farmland, perhaps due to public scrutiny (Barbarino 2017). Emerging research shows that large-scale corporate farms, which often have roots in family farming, are furthering land consolidation. This process has been theorized as a ‘long land grab’, where settler-capitalist elites gradually acquire large landholdings over generations (Kelly-Bisson 2024). Some of these farms undergo a transition from ‘farm to firm’, taking on the organizational and business forms of large capitalist enterprises, as in the case of Canadian mega-farms with land holdings of more than 30,000 acres (Mangan et al. 2023).

Digital technologies have played a crucial role in reaching the massive geographic scale of these farms by providing the tools to visualize landholdings through satellite imagery and sensors. GPS guidance alleviates the need for machinery operators to be familiar with the fields they are seeding and

harvesting. With the ability to manage extensive amounts of land unlocked by digital agriculture, farmers seek to maximize the utility of these tools and to increase their own financial assets by expanding their land base, thus fuelling intense farmland concentration. Whereas institutional investors may acquire and operate parcels in geographically diverse regions to diversify their portfolios, corporate family farms are more likely to acquire as much land as possible in a concentrated region for management purposes. This type of farm consolidation and land concentration then has much more profound effects on the hollowing out of rural areas. The reconfiguration of labour through digitalization requires more scholarly attention, particularly as it relates to corporate family farms and mega-farms. Here, it is important to highlight class relations, such as the distinct effects of digital technologies on farmers and farm workers (Ogunyiola et al. 2025; Duncan 2025). Nuance is needed in understanding digital agriculture’s effects on labour. Not all labour will simply be replaced by digital tools and robotics, but rather, ‘cyborg farmers’ may continue to place emphasis on embodied knowledge and their agroecological contexts rather than simply accepting algorithmic rationalities of farm management platforms (Van der Velden et al. 2024).

3.2 | Land and Labour in the Context of a Changing Climate

The dynamics of agrarian change cannot be discussed without recognition of the changing climate. Climate change has restructuring effects on labour as well as land—what is being produced and where it is possible to produce (Paprocki et al. 2025). The extreme effects of climate change will be proportionately felt by farm labour engaged in outdoor manual work. Climate change, for instance, is likely to drive automation and robotics in agrarian settings as humans may not be able to do such work in extreme conditions (Comi et al. 2023).

As for the effects on land, climate change allows for expanding modes of extractivism as capital seeks new locations for investment in agriculture (Rasmussen and Lund 2018). For instance, institutional investors have been promoting Canada’s northern regions as a new frontier for commodity crop production, such as soybeans, due to warming. Land in this region is just beginning to be assembled for the process of assetization (Li 2015). These lands are the traditional territory of several Indigenous nations protected by treaty rights, raising crucial political questions about development (Rotz et al. 2025). While these processes are still unfolding, this example highlights the need to understand patterns of resistance and contestation to the intersecting forms of capitalism and colonization that characterize these lands as ‘new frontiers’ (Price et al. 2022; Price 2022). As some areas will prove to be less productive in the context of climate change or prone to higher risks from extreme events, institutional investors may withdraw from sites of investment as well. Monitoring how the climate crisis, as well as other geopolitical conflicts, alters the geographies of farmland assetization will be critical for future research. Scholars should seek to understand the logics of finance that lead institutional investors to take interest in new geographies, whether from climatological or political shifts, as well as the effects these processes have on local communities.

The links between climate change and financialization also shape what is produced on the land. Emerging research demonstrates that farmland investment firms are increasingly turning to permanent crops, such as orchards for fruit or nut-tree production, rather than row crops. Institutional investors have the capital to establish new permanent crop production, which takes several years before returns from crops materialize. Once a new orchard or vineyard is established, the returns are much higher than row crops. Access to ecological resources, such as water for irrigation, is an influential factor for determining farmland suitable for this type of investment and may be a limiting factor due to climate change. Another limiting factor for permanent crop production is labour, leading to hype around robotics and other digital tools for use in the permanent crop sector, which may further fuel the transition of farmland investments to permanent crop production (Legun and Birch 2021). More information on these types of farm management changes would provide a better understanding of the environmental effects of farmland assetization. Overall, contextual specificities, with attention to geography and history, are necessary despite the global effects of climate change and the international flows of agricultural finance and investment.

3.3 | Forms of Contestation and Future Directions

In thinking about the entanglements of finance, labour and digital technologies, we would do well to remember, as Alistair Fraser (2022) puts it, that ‘you can’t eat data’. The global production of food requires land and labour, not the cloud, to produce it, ultimately leading to resistance to the ways that technology foregrounds the dynamics of finance, land and labour. Forms of contestation to these processes continue to emerge, arguing for the need for responsible innovation and forms of emancipatory smart farming (Bronson 2019; Jakku et al. 2023; Fraser 2022). These points of resistance manifest in organizing around the right-to-repair for farm machinery (Carolan 2020), advocating for increased interoperability and more open-source solutions for farmers (Rotz et al. 2019) and better mechanisms for agricultural data governance (Duncan 2023). Other possibilities include farmer cooperatives that manage agricultural data, which could prevent the use of agricultural data for speculation on commodity and farmland markets without the express consent of farmers. Additionally, farmers may simply not adopt digital agricultural tools that make them vulnerable to forms of agrarian surveillance capitalism, especially when they are uncertain about the return on investment of costly technologies (Abdulai et al. 2024).

The shifting terrain outlined above requires new conceptualizations, but it also raises methodological challenges for agrarian scholars. It is difficult to study the often-opaque processes of financialization, such as tracking land deals. While social science methods of discourse analysis have led to key insights (Visser 2017; Duncan and Magnan 2023), it is increasingly difficult to access public land registry data that can be used to provide information on the activities of farmland investors, in part because of the privatization of land data (Funk et al. 2025). Land registries are a vital data source for understanding questions of financialization and land within context-specific scales (Pritchard et al. 2023). Thus, we must be attentive not only to the

politics of land and the politics of (agricultural) data, but also the politics of accessing land data.

As critical agrarian scholars, we must continue to analyse the problematics of digital agriculture and its ties to financialization; however, we must also pay attention to how technologies could be co-opted for the purpose of resistance. Are there ways that these technologies might enable locally embedded food systems? What are the possibilities for technologies to be developed in ways that promote more equitable and dignified work opportunities in rural communities? While digital agriculture has been overhyped as a solution to the most pressing problems facing global agricultural production and has deepened inequalities, these are tools with a range of affordances (Duncan 2023). To advance this debate, there is a need to find opportunities to develop technologies that aid in resistance to the forces of financialization and promote desired outcomes of ecological sustainability and social equity in agricultural production.

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4 | Commentary: Carla Gras

The contributions to this Forum provide valuable insight for centering finance within agrarian change. They do this by offering grounds to analyse four interrelated processes: (1) the distinct logic of the assetization of farmland and agricultural operations; (2) the variety of property structures to control land and other elements of agricultural production; (3) the mutually reinforcing ties between assetization, digital technologies and the development of new sites for financial investment in the agricultural sector (i.e., the creation of monetized exchanges involving data and expert knowledge, giving place for new types of service-provision); and (4) the impacts on the demand for labour and labour relations driven by shareholder value. To connect the above contributions with a broader analysis of agrarian change, I synthesize these collective insights with Bernstein’s key questions of agrarian political economy (see Bernstein 2010).

4.1 | Who Owns What?

Financial investment in farmland has restructured property relations and led in many places to increased land concentration. As pointed by Frederico, the assetization of landownership stands on an ‘architecture of invisibility’ wherein the inherent difficulties for tracking shareholder identities and sources of capital are deepened through the building of networks of subsidiaries and controlled firms, including local landowners and farmers. This architecture of invisibility therefore conceals landownership not only through legal devices but through the control of land owned by third-parties as well. Strictly speaking, the latter does not necessarily imply the selling of the whole landholding to investment funds but rather a share of the property, thus allowing investors to have strategic control, while the original landowner maintains ownership. This, in turn, facilitates total or partial exit strategies when needed.

Agricultural production alongside related nodes in agricultural value chains is also targeted by financial investors. The former might be closely tied to the maximization of shareholder value in farmland as, particularly in frontier spaces, the potential for profits—a condition that assets are required to fulfil (Visser 2017)—ultimately depend on agricultural operation for the appreciation of farmland prices (Fairbairn 2014; Clapp and Isakson 2018; Sosa Varrotti and Gras 2021). The own-operating and lease-out schemes distinguished by Fairbairn (2014) entail different types of direct and indirect control over working capital, infrastructure, machinery, logistics, etc. My own research in the soybean complex in Argentina identified investments by pension and equity funds in agribusiness corporations and seed companies, through which control of land and/or capital operation is achieved (Gras and Hernández 2016; Sosa Varrotti and Gras 2021). Duncan's contribution on the ownership of data on which AgTech and FinTech developments stand shows how financialization drives new instances of assetization and forms of property over resources that are relevant for agricultural operation and market exchange.

As a result, the dynamics of financialization have important implications for property regimes and compel us to pay attention to how control over land, capital and other key resources is achieved through the acquisition of all or part of a landholding or agricultural firm. The complexity of property structures accompanying assetization reveals the multiple relations of property and power that inhabit circuits of capital. These cannot be solely attributed to financialization but have been deepened by it. Importantly, this process raises questions about how agrarian capitals are being integrated and with what consequences, thus challenging agrarian studies to attend to issues related to the differentiation of classes of agrarian capital.

In the same vein, where financial investments in farmland and agricultural operations are tied to speculative and/or rent-seeking purposes, decisions as to how those objectives are pursued and through which specific mechanisms are transferred into the hands of managers (Frederico and Ouma). This entails the separation between the owners of capital (shareholders) and those in charge of the direction of the operation of capital. Identifying the class position of this group (Ouma) is equally challenging. While it has been discussed in the work of authors such as Poulantzas or E.O. Wright, it has not yet received specific attention in agrarian studies.

4.2 | Who Does What?

Both Ouma and Duncan point to how financial objectives in agricultural operations can be linked to novel forms of labour control and monitoring. The impacts of this transformation on the exploitation of labour demand closer examination. As they show, digital technologies have, among other things, enabled sophisticated tools for measuring and monitoring the labour process and pace of work. Critical for providing detailed metrics to shareholders, these mechanisms are also deployed in farmland leasing-out schemes and outsourced work in own-operated farmland (i.e., harvesting and sowing).

These contributions thereby pose a series of questions on the conditions in which labour is included in financialized agriculture. I highlight three considerations that I believe need addressing. Firstly, to what extent does this process foster differentiation between types of tasks and the workers appointed to them, and how is this reflected in different working and employment conditions? Secondly, to what extent does this reshape and generate new lines of differentiation—stratification—within workers in on-farm tasks as well as between them and those working in outsourced tasks? Thirdly, in what ways and to what extent are such segmentation processes within the labour force adequately explored by existing categories of formal/informal/precarious labour?

These developments are not limited to assetized agricultural operations. Investments in agribusiness and large-scale farming provide examples where production, market and financial objectives are largely intertwined as shown in the case of large-scale commodity producers in Argentina (Sosa Varrotti and Gras 2021; Gras and Hernández 2016). In this regard, a relevant theme pertains to the role of finance in contract farming schemes, and more broadly, the proliferation of service companies linked to agribusiness and large-scale farming firms. To make rapid profits over one agricultural season, 'associated harvests' schemes are commanded by these actors in Argentina and integrated by capitalist farmers of different sizes to channel investments from different sources of capital including industrial corporations, energy distribution companies and domestic savings from rich urban classes. They are promoted as a means of sustaining farming activities through a more 'reasonable' risk exposure—particularly in times of market volatility and/or rocketing land lease prices. Associated harvests establish what each participant contributes and what they get. The precise arrangements vary and include combinations of elements usually attributed to service and production contracts. These schemes have also allowed large-scale agricultural firms to manage risks associated with diseconomies of scale while offering flexible investment options for investors. Other examples are the locally called 'management services' offered by some agribusiness corporations through a network of business units and property-controlled firms. These units provide a variety of services extending from the organization of the production process (aided by precision AgTechs) to the origination of grain⁶—allowing the management company to either purchase at pre-fixed prices (a type of contract called forward)⁷, or use derivatives such as future market contracts (Gras 2025). Managing these different options demands sophisticated financial knowledge. Important income differences can result during this process, which allows investors the possibility of benefitting from these business models. Importantly, these schemes do not incorporate the labour of low-cost self-exploited farmers, but agrarian capitals of different sizes. Through such schemes, large capitals are able to influence/shape the strategies of accumulation of smaller capitals. This opens new avenues for the examination of concentration in agriculture and how it is shaped by finance.

4.3 | Who Gets What and What Do They Do With It?

The assetization of farmland and agricultural operations is also reshaping how value is distributed between capital and labour, although this remains, to a large extent, an open topic

of research. Issues related to the appreciation of shares held by financial investors, management and performance fees established for asset managers come together when assets are resold, and capital gains are realized. As underlined by Braun (2025) capital gains are a key factor driving income and wealth inequality; focusing on the United States, Braun shows that capital gains have experienced remarkable growth compared to other income sources and are highly concentrated among the richest 10% of the population. Private equity holdings, including those linked to agriculture, emerge as major drivers of this substantial shift in income inequality. This is a key manifestation of the empowered role of finance in contemporary capitalism.

Within agrarian studies, paying attention to how rent-seeking logics operate entails considering how returns on financial investments are not only pursued but also remunerated and, importantly, the extent to which they further increase capital profits to the detriment of workers (as suggested by Cochet 2018). Alongside this fundamental tension between capital and labour, we also need to consider the potential for accompanying disputes between productive and financial capitals, and therefore how these dynamics are recomposing contemporary accumulation processes.

To this end, the contributions by our guest authors provide insightful analytical frameworks to better connect the role of finance and financial mechanisms to changing dynamics of accumulation, property and power relations and class relations within contemporary capitalism. At the same time, grounded on many years of empirical research, the work of Frederico, Ouma and Duncan also provides important methodological grounds from which to integrate finance fruitfully into the active research agenda of agrarian change, as attempted in these comments.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

Endnotes

¹ See Gras, C., Sinha, S., Zhang, F., Jansen, K., Luna, J. and Taylor, M. (2025).

² <https://farmlandgrab.org/post/25207-tiaa-cref-unveils-3-billion-global-ag-investment-vehicle#sthash.4WAinG4L.dpuf>.

³ <https://institutionallandscapes.org/follow-the-money/>.

⁴ I deploy the notion of ‘capital placements’ rather than ‘investments’: ‘Placement means the purchase of titles to debts or shares, which is financed either from savings, from income or from the proceeds of selling other property. In contrast, investment designates using financial resources for creating capital goods’ (Robinson 1956, 8; cited in Zeller 2008, 10).

⁵ <https://mapof.ag/>.

⁶ In the trading business jargon, origination is the first stage of commercialization: It entails sourcing and delivering grains to local warehouses for final dispatch to exporters.

⁷ <https://www.investopedia.com/ask/answers/06/forwardsandfutures.asp>.

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